Homework 1

The Capstone Challenge: The "Smart Studio Lamp"

The Scenario: You have been hired to prototype a control panel for a photographer's studio light. The photographer needs to turn the light on and off with a single button press. When the light is on, they need a dial to precisely adjust the brightness. They also want a digital readout of the light's current power level on their computer.

Hardware Required:

- 1x Push Button (with 10kΩ pull-down resistor)
- 1x Potentiometer
- 1x LED (with appropriate current-limiting resistor)
- Jumper wires and Breadboard

The Functional Requirements (The Rules):

1. **The Master Toggle:** The push button must act as a master ON/OFF switch. Pressing it once turns the system ON. Pressing it again turns the system OFF. (Requires state change logic and debouncing from Week 2).
2. **Adjustable Brightness:** When the system is ON, the LED's brightness must be controlled in real-time by turning the potentiometer. (Requires Analog Reading and PWM from Week 3).
3. **True Zero:** When the system is OFF, the LED must be completely dark, regardless of where the potentiometer dial is currently turned.
4. **Smart Serial Telemetry:** The Serial Monitor must display the status of the lamp.
    - If the lamp is OFF, it should print: `System OFF`.
    - If the lamp is ON, it should print the brightness level.
    - *The Catch:* Do not spam the Serial Monitor! The Arduino should **only** print a new line when the button state changes OR when the potentiometer dial is actually moved.

Homework 2

The Capstone Challenge Part II: The "Industrial Pressure Monitor"

The Scenario: You are designing a safety monitoring system for a high-pressure steam valve. The potentiometer represents a pressure sensor. The LED is the warning beacon on the operator's dashboard, and the button is the "Acknowledge Alarm" switch.

The Functional Requirements (The Rules):

1. **Zone 1: The Safe Zone (Analog reading 0 – 511)**
    - The pressure is normal. The LED must be completely OFF.
2. **Zone 2: The Warning Zone (Analog reading 512 – 900)**
    - The pressure is rising. The LED must turn ON, and its brightness must scale with the dial.
    - *The Catch:* At an analog reading of 512, the LED should be barely on (PWM ~1). At a reading of 900, it should be at maximum brightness (PWM 255).
3. **Zone 3: The Critical Zone (Analog reading 901 – 1023)**
    - The pressure is dangerously high. The LED must detach from the PWM brightness control and instead **blink** continuously.
4. **The "Acknowledge" Override**
    - If the system is in the Critical Zone (blinking), the operator can press the push button to acknowledge the danger.
    - Once pressed, the LED stops blinking and stays solid ON at maximum brightness.
    - If the dial is turned back down below 901, the system resets to normal behavior, and the override is forgotten.
5. **Serial Monitor Output**
    - Continuously print the system status using a clear format. Example: `Status: WARNING | Sensor: 750 | PWM: 154`

The Final Boss: The "Engine Control Unit (ECU) Simulator"

The Scenario: You are programming the logic for a simplified electronic vehicle transmission. The potentiometer acts as the gas pedal (throttle). The push button acts as the gear shifter. The LED represents the engine RPM/stress.

The Functional Requirements (The Rules):

You must create a program with **four distinct states**. The system starts in State 0.

- **State 0: Engine OFF**
  - The LED is completely OFF.
  - The potentiometer does absolutely nothing.
  - *Action:* Pressing the button shifts the system to State 1.
- **State 1: Engine IDLE (Neutral Gear)**
  - The engine is running, but you aren't in gear.
  - The LED must continuously and smoothly pulse up and down (like the "Sunrise Simulator" from your Week 3 homework, fading from 0 to 255 and back down).
  - The potentiometer still does nothing.
  - *Action:* Pressing the button shifts the system to State 2.
- **State 2: First Gear (Standard Drive)**
  - The pulsing stops immediately.
  - Now, the LED brightness is directly controlled by the potentiometer (Throttle). 0 on the dial = 0 brightness. 1023 on the dial = 255 brightness.
  - *The Catch:* You are **locked** in First Gear. Pressing the button will do nothing *unless* the throttle is maxed out.
  - *Action:* If the potentiometer is reading greater than 1000 AND you press the button, you shift to State 3.
- **State 3: Reverse Gear (Inverted Drive)**
  - The system is in reverse. The logic is now flipped.
  - If the potentiometer is at 0, the LED is at MAXIMUM brightness. As you turn the dial up to 1023, the LED dims down to completely OFF.
  - *Action:* Pressing the button at any time in this state turns the engine OFF (Returns to State 0).